TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	77.7 ± 1.62
Correlation Stability	81.2 ± 1.83
Distribution Consistency	87.4 ± 1.50
Entropy Preservation	90.7 ± 1.79
Mean Z Score Drift	0.2 ± 0.02
Categorical Fidelity	87.6 ± 1.28
Privacy Risk	53.1 ± 3.27
Data Density	100.0 ± 0.01
Coverage Score	12.8 ± 3.34
Feature Accuracy	78.6 ± 3.26
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	claude-haiku-5-5
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	47.629	+2.882	6.4%	OK
TB	3.299	1.464	-1.835	55.6%	Drift
DB	1.486	0.587	-0.899	60.5%	Drift
Alkphos	290.576	222.327	-68.249	23.5%	Warn
Sgpt	80.714	40.452	-40.262	49.9%	Drift
Sgot	109.911	49.137	-60.774	55.3%	Drift
TP	6.483	6.643	+0.159	2.5%	OK
ALB	3.142	3.346	+0.204	6.5%	OK
A/G Ratio	0.947	1.009	+0.062	6.6%	OK
Selector	1.286	1.420	+0.133	10.4%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	77.7 ± 1.62
Correlation Stability (*2)	81.2 ± 1.83
Distribution Consistency (*3)	87.4 ± 1.50
Entropy Preservation (*4)	90.7 ± 1.79
Mean Z Score Drift (*5)	0.2 ± 0.02
Categorical Fidelity (*6)	87.6 ± 1.28
Privacy Risk (*7)	53.1 ± 3.27
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	12.8 ± 3.34

Feature Accuracy (*10)	78.6 ± 3.26
Novelty Score (*11)	0.4 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7042735042735042	0.6846153846153846	0.7111111111111111	0.00683760683760688	
F1	0.8074638483152219	0.7611255703238442	0.8057961226899263	-0.0016677256252955797	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6760683760683761	0.7136752136752137	0.011111111111111105	
F1	0.7997903571549948	0.75755405413696	0.8032621306471188	0.0034717734921242416	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6811965811965812	0.6888888888888889	-0.01709401709401708	
F1	0.8128002697439015	0.7605426576048571	0.7785677133256704	-0.03423255641823099	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6871794871794872	0.6931623931623931	0.01880341880341883	
F1	0.7821343746683586	0.7676295015077064	0.7803060750216275	-0.001828299646730908	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6871794871794872	0.6786324786324787	0.6965811965811965	0.009401709401709368	
F1	0.7950564352931886	0.7597940501708447	0.7841108187171628	-0.010945616576025729	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6803418803418804	0.6923076923076923	-0.014529914529914534	
F1	0.8279091624529537	0.7513285317181425	0.7763489836587768	-0.05156017879417695	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.6965811965811965	0.6230769230769231	0.06410256410256412	*
F1	0.5587432119380917	0.7768008345022632	0.6586390179547732	0.09989580601668153	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6581196581196582	0.6145299145299146	0.6923076923076923	0.0341880341880342	*
F1	0.7608910172750105	0.7029991498407037	0.7801948366385005	0.01930381936349016	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	0.6512820512820513	0.7128205128205128	0.028205128205128216	*
F1	0.7903491772809568	0.7289570910193734	0.8014855469124038	0.011136369631447051	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6854700854700855	0.6948717948717948	0.00854700854700854	
F1	0.7971299319768493	0.760832759422421	0.7821357776023835	-0.014994154374465895	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	70	Male	1.7	0.5	400	56	44	5.7	3.1	1.1
2	45	Male	2.5	1.2	163	28	22	7.6	4.0	1.1
3	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
4	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
5	45	Male	0.8	0.2	140	24	20	6.3	3.2	1.0
6	42	Female	0.8	0.2	195	18	15	6.7	3.0	0.8
7	58	Male	1.2	0.4	210	34	41	6.9	3.3	0.9
8	44	Female	0.9	0.2	182	22	19	7.1	3.6	1.0
9	67	Male	3.4	1.7	298	52	77	6.2	2.7	0.8
10	36	Male	0.7	0.2	164	19	23	7.4	4.1	1.2
11	52	Female	1.1	0.4	201	28	31	6.5	3.2	0.9
12	73	Male	0.8	0.2	225	26	29	5.6	2.6	0.9
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	16	Male	0.7	0.2	418	28	35	7.2	4.1	1.3
2	36	Male	0.8	0.2	158	29	39	6.0	2.2	0.5
3	14	Male	0.9	0.3	310	21	16	8.1	4.2	1.0
4	48	Female	1.4	0.8	621	110	176	7.2	3.9	1.1
5	17	Female	0.5	0.1	206	28	21	7.1	4.5	1.7
6	61	Male	0.8	0.2	192	28	35	6.9	3.4	0.9
7	45	Male	1.2	0.4	215	34	41	6.8	3.2	0.9
8	29	Female	0.7	0.2	176	22	25	7.1	4.0	1.3
9	63	Male	3.4	1.7	402	58	96	6.3	2.8	0.8
10	52	Female	0.9	0.3	238	31	36	6.5	3.4	1.1
11	34	Male	0.8	0.2	189	27	30	7.4	4.1	1.2
12	70	Male	1.6	0.6	295	44	52	5.9	2.6	0.8
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	1.0	0.3	154	38	21	6.8	3.9	1.3
2	21	Male	18.5	9.5	380	390	500	8.2	4.1	1.0
3	38	Male	0.8	0.2	208	25	50	7.1	3.7	1.0
4	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
5	55	Male	0.6	0.2	220	24	32	5.1	2.4	0.88
6	32	Male	30.5	17.1	218	39	79	5.5	2.7	0.9
7	45	Male	1.2	0.4	186	34	41	6.9	3.5	1.0
8	29	Female	0.7	0.2	165	22	25	7.2	3.9	1.2
9	61	Male	2.1	1.0	240	58	72	6.2	2.9	0.9
10	34	Female	0.9	0.3	152	18	20	6.6	3.4	1.1
11	52	Male	0.8	0.2	198	27	33	7.0	3.6	1.1
12	67	Female	1.4	0.6	274	46	58	5.8	2.6	0.8
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	37	Male	0.8	0.2	147	27	46	5.0	2.5	1.0
2	65	Male	7.9	4.3	282	50	72	6.0	3.0	1.0
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	42	Female	7.4	3.6	298	52	102	4.6	1.9	0.7
5	75	Male	14.8	9.0	1020	71	42	5.3	2.2	0.7
6	69	Female	0.8	0.2	146	42	70	8.4	4.9	1.4
7	44	Male	1.2	0.4	215	32	41	6.8	3.4	1.0
8	58	Female	0.9	0.2	174	22	28	7.1	3.6	1.0
9	63	Male	3.4	1.6	312	55	68	6.2	2.7	0.8
10	29	Female	0.7	0.1	142	18	21	7.6	4.1	1.2
11	71	Male	5.6	2.7	480	88	115	5.9	2.6	0.8
12	36	Male	0.8	0.2	165	30	34	6.9	3.5	1.0
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	1.7	1.0	180	18	34	7.2	3.6	1.0
2	18	Male	1.4	0.6	215	440	850	5.0	1.9	0.6
3	22	Male	2.7	1.0	160	82	127	5.5	3.1	1.2
4	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
5	33	Male	0.7	0.1	168	35	33	7.0	3.7	1.1
6	62	Male	5.0	2.1	103	18	40	5.0	2.1	1.72
7	44	Male	1.2	0.4	190	32	41	6.8	3.4	1.0
8	29	Female	0.9	0.2	175	22	25	7.2	3.9	1.2
9	61	Male	3.1	1.5	260	58	77	5.9	2.7	0.8
10	55	Female	0.7	0.2	210	18	21	6.5	3.2	1.0
11	33	Male	1.5	0.6	230	70	95	6.1	3.0	0.9
12	68	Male	0.8	0.2	160	24	30	7.4	4.0	1.2
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39	Male	1.6	0.8	230	88	74	8.0	4.0	1.0
2	32	Male	18.0	8.2	298	1250	1050	5.4	2.6	0.9
3	48	Male	2.4	1.1	554	141	73	7.5	3.6	0.9
4	41	Female	0.9	0.2	201	31	24	7.6	3.8	1.0
5	42	Male	0.8	0.2	158	27	23	6.7	3.1	0.8
6	60	Female	1.4	0.7	159	10	12	4.9	2.5	1.0
7	45	Male	1.2	0.4	210	34	40	6.8	3.5	1.0
8	62	Female	0.9	0.2	175	22	27	7.1	3.8	1.1
9	35	Male	2.8	1.3	260	95	88	6.2	3.0	0.9
10	58	Male	0.7	0.2	142	18	21	6.9	3.7	1.2
11	29	Female	0.8	0.2	190	26	24	7.6	4.0	1.1
12	71	Male	1.5	0.6	310	45	62	5.5	2.6	0.9
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Female	0.7	0.2	164	21	53	4.5	1.4	0.45
2	50	Male	7.3	3.6	1580	88	64	5.6	2.3	0.6
3	70	Male	2.7	1.2	365	62	55	6.0	2.4	0.6
4	45	Female	0.7	0.2	153	41	42	4.5	2.2	0.9
5	55	Male	1.8	9.0	272	22	79	6.1	2.7	0.7
6	38	Male	1.5	0.4	298	60	103	6.0	3.0	1.0
7	44	Male	1.2	0.4	198	34	41	6.8	3.4	1.0
8	58	Female	0.9	0.2	176	25	29	6.5	3.2	0.96
9	62	Male	3.1	1.4	240	58	77	5.9	2.6	0.8
10	27	Female	0.7	0.2	155	19	22	7.4	4.0	1.18
11	39	Male	0.8	0.2	182	28	33	6.9	3.5	1.03
12	71	Male	1.6	0.6	310	46	52	6.1	2.8	0.85
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Female	0.8	0.2	168	25	18	6.2	3.1	1.0
2	52	Male	0.6	0.1	171	22	16	6.6	3.6	1.2
3	52	Male	2.7	1.4	251	20	40	6.0	1.7	0.39
4	51	Female	0.9	0.2	280	21	30	6.7	3.2	0.8
5	55	Male	3.6	1.6	349	40	70	7.2	2.9	0.6
6	35	Male	2.0	1.1	226	33	135	6.0	2.7	0.8
7	45	Male	1.1	0.3	210	28	35	6.8	3.4	1.0
8	62	Female	0.9	0.2	245	19	24	6.4	3.1	0.9
9	38	Male	2.4	1.1	290	64	92	7.1	3.6	1.0
10	29	Female	0.7	0.2	175	22	20	7.5	4.2	1.2
11	57	Male	4.6	2.3	330	88	152	6.1	2.6	0.7
12	33	Male	0.8	0.2	188	31	29	7.2	3.8	1.1
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	27	Male	1.0	0.2	205	137	145	6.0	3.0	1.0
2	58	Male	1.0	0.5	158	37	43	7.2	3.6	1.0
3	38	Male	0.9	0.3	310	15	25	5.5	2.7	1.0
4	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
5	31	Male	0.6	0.1	175	48	34	6.0	3.7	1.6
6	51	Male	0.8	0.2	175	48	22	8.1	4.6	1.3
7	45	Male	1.2	0.4	210	35	42	6.8	3.4	1.0
8	62	Female	0.9	0.2	175	28	31	7.1	3.8	1.1
9	34	Male	2.8	1.3	245	95	110	6.2	3.0	0.9
10	58	Female	0.7	0.2	160	22	25	7.4	4.0	1.2
11	29	Male	0.8	0.2	188	41	36	6.5	3.5	1.2
12	71	Male	1.6	0.7	290	48	66	6.0	2.9	0.9
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Female	0.7	0.2	152	35	81	6.2	3.2	1.06
2	34	Male	5.9	2.5	290	45	233	5.6	2.7	0.9
3	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
4	18	Male	0.7	0.1	312	308	405	6.9	3.7	1.1
5	36	Male	0.8	0.2	182	31	34	6.4	3.8	1.4
6	26	Male	1.3	0.4	173	38	62	8.0	4.0	1.0
7	45	Male	1.2	0.4	210	38	44	6.9	3.4	0.9
8	38	Female	0.8	0.2	175	24	29	7.2	3.8	1.1
9	62	Male	3.1	1.5	320	72	98	6.4	2.8	0.8
10	29	Male	0.9	0.3	165	41	36	7.6	4.1	1.2
11	51	Female	0.7	0.2	198	19	27	6.8	3.5	1.0
12	70	Male	1.6	0.6	260	33	41	5.8	2.6	0.8
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3